Schumann Konen

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EDUCATION

The University of Tennessee, Knoxville, Tickle College of Engineering **Knoxville, TN**
Bachelor of Science in Industrial and Systems Engineering Minor in BM. 2024-Dec. 2026
Pellissippi Community College **Knoxville, TN**
Associate of Applied Science in Mechanical Engineering 2022–2024

PROJECT EXPERIENCE

University of Tennessee **Knoxville, TN**
Senior Design Project — Nonprofit Data & Process Improvement 2022-Present

Industrial & Systems Engineering Capstone

• Analyzed organizational data to identify improvement opportunities in tracking, decision-making, and reporting for nonprofit operations.
• Developed decision trees and structured data processes to help the organization better understand recipient outcomes and funding impact.
• Designed volunteer surveys and data collection methods to improve feedback quality, operational visibility, and future decision-making.
• Connected Industrial Engineering tools with real-world service operations by improving how the nonprofit organizes information and measures impact.

Manufacturing Design Class Project (Pen Factory)

• Led a semester-long IE401 manufacturing flow and facility-layout improvement effort by mapping end-to-end routing, quantifying baseline improving 25% flow efficiency, and redesigning department adjacencies to cut transport/double-handling and improve projected flow efficiency to 50%
• Designed the Layout of the facility: shipping/receiving area, storage, staging, work/processing areas.
• Mapped material flow from receiving, processing, storage, staging to shipping, identify bottlenecks and constraints to reduce nonvalue added movement.

Hybrid AnyLogic Simulation Project

• Developed a hybrid casino simulation model in AnyLogic using both discrete-event simulation and agent-based modeling to represent customer flow, gaming activity, resource usage, and Casino Profits.
• Defined the simulation scope by modeling customer arrivals, seasonal demand patterns, game selection, queue formation, table assignment, dealer/resource availability, gameplay duration, customer movement, and system exit behavior.
• Applied forecasting and seasonality concepts to represent changes in customer demand across different time periods, allowing the model to test how peak and non-peak arrival patterns affected wait time, congestion, and throughput.
• Utilized operations research principles to evaluate resource allocation decisions, including dealer staffing levels, table capacity, queue behavior, utilization rates, bottlenecks, and throughput under multiple operating scenarios.
• Analyzed simulation outputs including queue length, wait time, throughput, table/dealer utilization, and total time in system to recommend improvements for casino flow and operational efficiency.

Hydropower Hammer Project

• Collaborated with a team of four to design a hydropower hammer. Utilized fluid mechanics, torque, and rotational motion for a successful design, maximized durability and efficiency by 50% using newton laws.
• Researched numerous materials to construct a waterwheel for durability and cost efficiency, to avoid excessive spending.
• Utilized AutoCAD for 3D designs, constructing the basic structures of our design.

WORK EXPERIENCE

DormCo. **Nashville, TN**
Supply Chain/Warehouse Operations Intern May 2026
• Supported daily e-commerce fulfillment operations by assisting with picking, packing, shipping, and warehouse organization for college dorm product orders.
• Improved understanding of supply chain execution by working hands-on with order flow, inventory handling, shipment preparation, and high-volume seasonal demand.
• Collaborated with warehouse staff to maintain efficient workflow during peak summer operations, including overtime periods of increased order volume.
• Applied Industrial & Systems Engineering concepts to observe bottlenecks, labor flow, space utilization, and opportunities for process improvement within fulfillment operations.

The Chop House **Knoxville, TN**
Server *2025*
• Provided friendly, professional service in a fast-paced restaurant while making sure guests felt welcomed and taken care of throughout their meal.
• Managed multiple tables at once by taking orders, checking on guests, handling requests, and keeping service running smoothly during busy shifts.
• Communicated with kitchen staff and team members to help ensure orders were accurate, delivered on time, and handled with attention to detail.
• Built strong customer service, multitasking, and problem-solving skills by working directly with guests and responding quickly to their needs.

HONORS, SKILLS& INTERESTS

Computer: Excel, SQL, Tableaus, AnyLogic, Pulp, RStudio, Python, ANOVA, DOE, Lucidchart, MATLAB, Jupiter Notebook
Skills: Discrete Event Simulation, Value Stream Mapping, Regression Analysis, Process Mapping, Forecasting
Language: Fluent in Mandarin | Interests: Powerlifting, Running, Career Development, Personal Finance, Cooking